



Jersey City Bitcoin Socratic Seminar #5

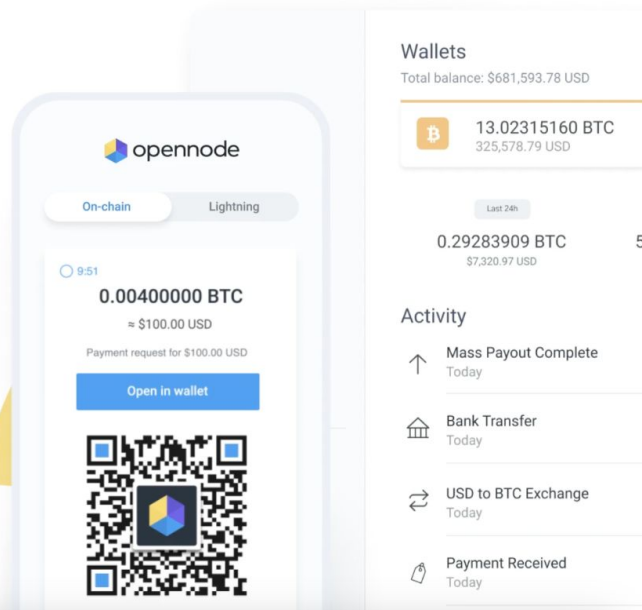
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Bitcoin payments at the speed of now

Get lightning-fast, low-cost bitcoin payments and payouts for your business with our powerful API, ecommerce plugins, or hosted payment pages.



Agenda:

- Club Leadership: Bitcoin Park trip + April marketing push
- Macro Update: War, Debt, Tariffs
- Bitcoin Price Update
- Podcast of the Month: Luke Gromen
- Bitcoin History: Cypherpunk Manifesto
- Freedom Tech Spotlight: White Noise
- Demo: White Noise live walkthrough
- Guest Session: Multisig
- What is Bark/Ark?



JC Bitcoin Meetup General Guidelines

- Basic Rules:

- Please no photos, no videos, no public tagging without consent!
- This group is Bitcoin only so please limit discussion of other cryptocurrencies.
- PLEASE respect the space. We love the barrow mansion and ask for help making it look better than when we arrived.
- For newcomers and OGs alike it is worth reiterating that absolutely no one from JC Bitcoin will ever solicit Bitcoin or crypto transfers here or on any other channel

Number Time

Clark Moody Bitcoin			Dashboard		
Markets ?			Blockchain ?		
Price	\$68,660		Block Height	936,815	
Sats per Dollar	1,456		UTX0 Set Size	164,297,540	
Market Capitalization	\$1.37T		Block Time, Prior Year	9:57	
Drop some sats in the Tip Jar			Chain Size	820.9 GB	
			OP_RETURN Data	0.0 GB	
All-Time High ?			Bitcoin Network ?		
All-Time High Price	\$126,160		Reachable Bitcoin Nodes	25,241	
Decline from ATH	-45.60%		Bitcoin Tor Nodes	16,484	
ATH Date	October 6, 2025		Percentage Tor Nodes	65.31%	
Days Since ATH	132		Lightning Network (Public) ?		
Gold ?			Total Capacity	4,997.61 BTC	
Bitcoin priced in Gold	16.3 oz		Capacity Value	\$343.1M	
Bitcoin vs Gold Market Cap	4.59%		Total Nodes	9,949	
Corporate Treasuries ?			Total Channels	39,262	
Held in Corp. Treasuries	2,285,604 BTC		Tor Capacity	3,972.52 BTC	
Value in Corp. Treasuries	\$156.9B		Percentage Tor Capacity	79.5%	
Supply Pct. in Corp. Treasuries	11.43%		Tor Nodes	5,386	
Supply ?			Bitfinex Lightning Node ?		
Money Supply	19,989,819.91 BTC		Inbound Capacity	233.48 BTC	
Percentage Issued	95.19%		Outbound Capacity	857.30 BTC	
Unspendable	230.09 BTC		Bitfinex Channels	560	
Issuance Remaining	1,009,949.98 BTC		Liquid Sidechain ?		
			Peg-in Capacity	-96.06 BTC	
			Peg-in Capacity Value	\$-6,595,191.89	
			Liquid Block Height	3,758,944	
			Liquid Chain Size	52.0 GB	
			Transactions ?		
			Total All Time	1,310,843,708	
			Rate, 30 days	4.8 tx/s	
			Count, 30 days	12,742,529	
			Chain Security ?		
			Hash Rate, 90 Days	1,020.0 EH/s	
			Chain Work	96.0 bits	
			Chain Rewrite Days	896	
			Annual Mining Revenue	\$16.6B	
			Mining ?		
			Hash Rate, 2016 Blocks	1,023.2 EH/s	
			Difficulty	125.9x10 ¹²	
			Difficulty Epoch	465	
			Last Difficulty Change	-11.2%	
			Block Time, 2016 Blocks	9:09	
			Difficulty Retarget ?		
			Blocks to Retarget	625	
			Retarget Date	February 19, 2026	
			Estimated Difficulty Change	14.9%	
			Block Time, Diff. Epoch	8:42	
			Mining Economics ?		
			Block Subsidy	3.13 BTC	
			Block Subsidy Value	\$214,563	
			Daily PHash/s Revenue	48,374 sats	
			Daily PHash/s Value	\$33.67	
			Avg. Fees per Block	0.02 BTC	
			Avg. Fees vs. Reward	0.64%	

Quick Mempool Check







Explore the full Bitcoin ecosystem

~0 sat/vB
0.12 - 0.12 sat/vB
1 BTC
72 transactions
In ~34 minutes

~0 sat/vB
0.12 - 0.12 sat/vB
0.001 BTC
72 transactions
In ~26 minutes

~0 sat/vB
0.12 - 0.14 sat/vB
0.001 BTC
2,663 transactions
In ~17 minutes

~0 sat/vB
0.14 - 403 sat/vB
0.009 BTC
4,656 transactions
In ~8 minutes

936815
~1 sat/vB
0.13 - 244 sat/vB
0.019 BTC
3,635 transactions
7 minutes ago
MARA Pool

936814
~0 sat/vB
0.10 - 5.00 sat/vB
0.00 BTC
2,373 transactions
27 minutes ago
SECPool

936813
~0 sat/vB
0.12 - 224 sat/vB
0.009 BTC
1,887 transactions
27 minutes ago
MARA Pool

936812
~1 sat/vB
0.13 - 212 sat/vB
0.012 BTC
2,879 transactions
33 minutes ago
AntPool

TRANSACTION FEES

No Priority	Low Priority	Medium Priority	High Priority
0.1 sat/vB	0.1 sat/vB	0.5 sat/vB	1 sat/vB
\$0.01	\$0.01	\$0.05	\$0.10

DIFFICULTY ADJUSTMENT

difficulty halving

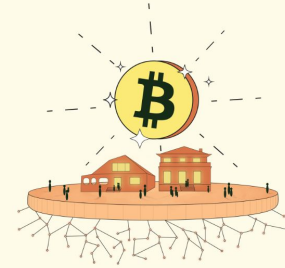
~8.7 minutes
Average block time

▲ 14.93 %
Previous: ▼ 11.16 %

In ~3 days
February 19 at 1:31 PM

Bitcoin Park & Grassroots Summit

- Goal: bring back best practices, connections, and resources for JC Bitcoin
- Grassroots Quarterly Roundtable
- DISCUSS:
 - What do you want to learn?
 - What should I ask the meetup organizers?
- What if we had an initiative to circulate btc
- Bulletin board of services, really start the circular economy



GRASSROOTS
BITCOIN 2026



Marketing Initiatives

- Challenge: every member brings ONE friend
- Marketing meetings - Sunday Coffee shop or zoom?
- Openclaw?
- Can someone print stickers/Fliers?
 - Bulletin board
 - Fliers at Helen's
- Fundraiser - Networking Event
- All about Downtown 9/19
 - Big Orange Tent ~\$400
- \$50 Yearly Membership - Satoshi Fellow
 - Resume builder
 - Merch!
 - Members Group Chat Access



Macro Update

- War is disrupting supply chains
 - LNG, Oil
 - Fertilizer
 - Helium - Chips
- “Feels like February 2020”
- National debt reaches \$39 trillion
- Ai deflation and the big print
- Private Credit issues getting worse
 - Gating funds, classic late cycle red flag
- Dollar - DXY increasing
- Ai Capex Spend decreases liquidity
 - Global liquidity likely peaked
- **Simultaneous shocks to an overleveraged system**



laurence 
@functi0nZer0

X.com

Claude, repair the LNG facilities before market open, make no mistakes

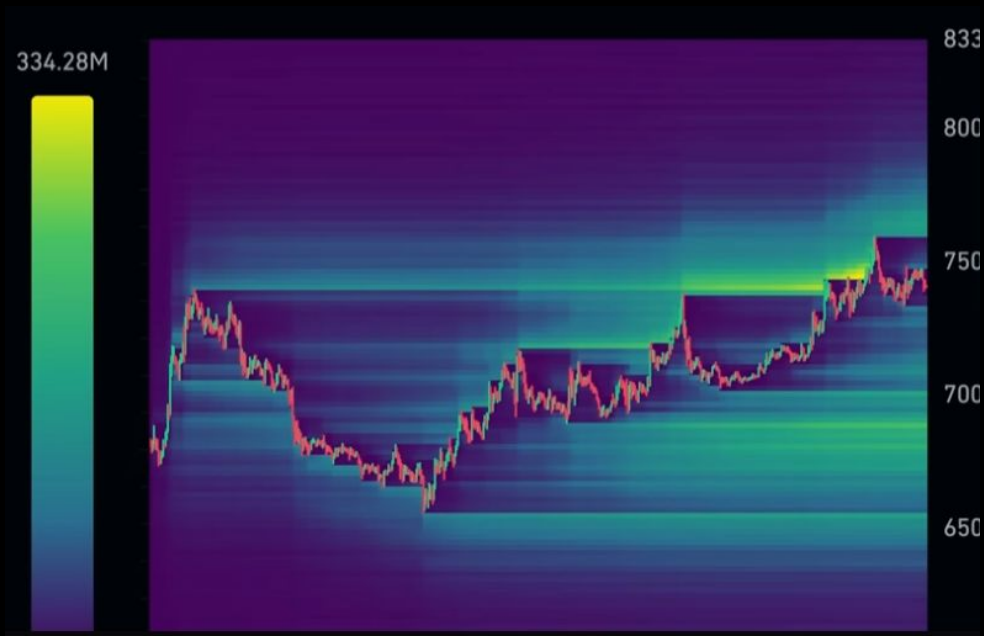
02:36 · 3/19/26 · **540K** Views

Bitcoin Price Update

- ETF flows flipping positive: +\$568M net in early March
- Strategy putting in some large purchases
- Bitcoin front ran and stocks lagged
- Playing out a bullish relief, but that doesn't mean the bottom is in
- HODL on!



Bitcoin Price Update



Bitcoin Price Update



Podcast Spotlight

Luke Gromen

- The big print is coming...very soon
- End of 100-year debt cycle + AI deflation
 - = sovereign debt crisis within **6-24 months**
- AI is deflationary, great for humanity, catastrophic for a debt-based system
 - Even 5-10% displacement cascades through a leveraged system
- February 2026 is the new July 2007
- Endgame: massive debasement, medium/long-term extremely bullish
- Important to make an action plan ASAP



Bitcoin History:

The Cypherpunk Manifesto

- Written by Eric Hughes, March 9, 1993
- "Privacy is necessary for an open society in the electronic age"
- *Cypherpunks write code*
 - They don't ask for permission
 - Doesn't have to be code, just action
- "Don't get too literal on us here. What is important is that Cypherpunks take personal responsibility for empowering themselves against threats to privacy."
 - CYPHERNOMICON 1994-09-10

A Cypherpunk's Manifesto

Eric Hughes

March 9, 1993

Privacy is necessary for an open society in the electronic age. Privacy is not secrecy. A private matter is something one doesn't want the whole world to know, but a secret matter is something one doesn't want anybody to know. Privacy is the power to selectively reveal oneself to the world.

If two parties have some sort of dealings, then each has a memory of their interaction. Each party can speak about their own memory of this; how could anyone prevent it? One could pass laws against it, but the freedom of speech, even more than privacy, is fundamental to an open society; we seek not to restrict any speech at all. If many parties speak together in the same forum, each can speak to all the others and aggregate together knowledge about individuals and other parties. The power of electronic communications has enabled such group speech, and it will not go away merely because we might want it to.

Since we desire privacy, we must ensure that each party to a transaction have knowledge only of that which is directly necessary for that transaction. Since any information can be spoken of, we must ensure that we reveal as little as possible. In most cases personal identity is not salient. When I purchase a magazine at a store and hand cash to the clerk, there is no need to know who I am. When I ask my electronic mail provider to send and receive messages, my provider need not know to whom I am speaking or what I am saying or what others are saying to me; my provider only need know how to get the message there and how much I owe them in fees. When my identity is revealed by the underlying mechanism of the transaction, I have no privacy. I cannot here selectively reveal myself; I must *always* reveal myself.

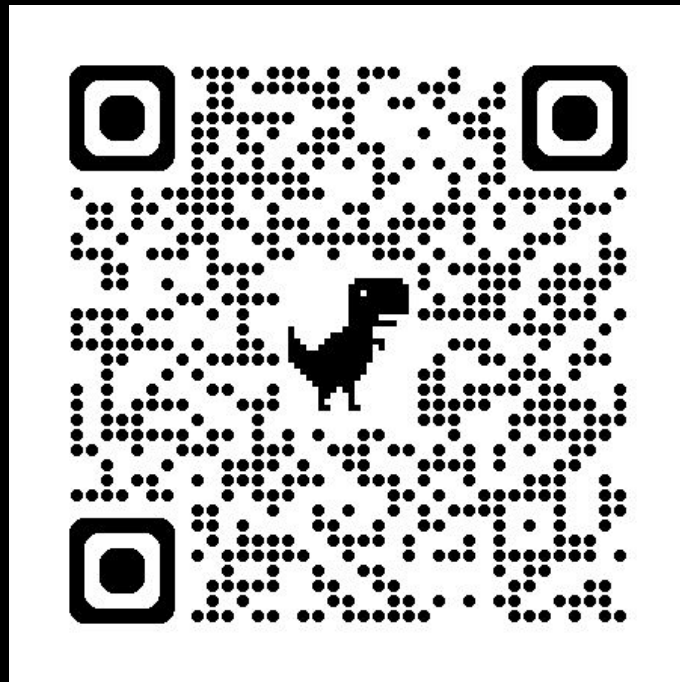
Freedom Tech Spotlight: White Noise

- Private encrypted messenger built on Nostr
- No phone number, no email, no central server
- Same encryption as Signal (MLS protocol) but decentralized
- "The chat with no phone number and no central server to shut down"



White Noise Demo

- Download: whitenoise.chat
- No signup -- just use your Nostr keypair (or create one)
- Available on: Apple TestFlight, Zapstore, Android APK
- Let's try it together!



Bitcoin Multi-Signature wallets for the ~~common man~~ pleb

Everything you don't understand
because you were afraid to find out



Syllabus

- Intro
- What (is it)
- Why (should it be used)
- How (is it used)
- When (should it be used)
- Where (should they be hidden)
- Who (should be involved)
- Whom (they are accessible to)
- Which (configuration to use)
- Hard & Soft (ware) Tips

Intro

- A **conceptual** not technical presentation
- Many avoid multi-sig because it sounds techy and intimidating, but in reality it is **just another tool** to protect your savings against the downsides of a single signature wallet.
- Common-sense safety, applied deliberately. It's for everyday people wanting **smarter security**.

FEARS

- “More keys = more chances to screw up”
 - **Reality: More keys = margin for error**
- “What if I lose one (or two)?”
 - **That’s literally what multi-sig is designed for**
- “Isn’t this for institutions?”
 - **Institutions adopted it because it works**
- “I don’t know how.”
 - **Technical setup is the same as Single-sig**

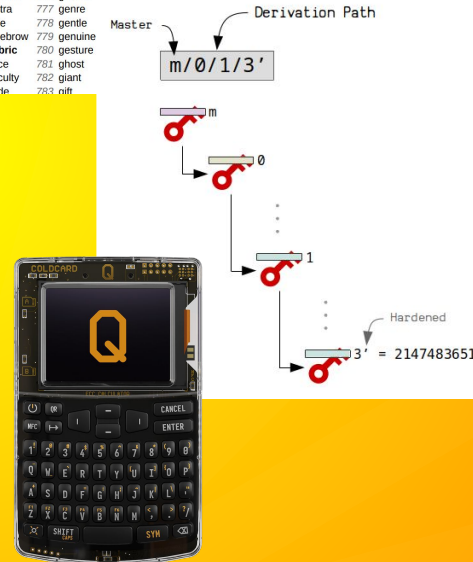
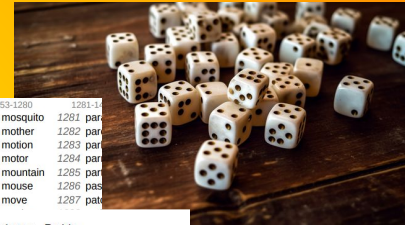
What is Multi-Sig

- What is a Single Signature wallet?
- Multi-Sig **builds on the Single signature** concept
- It uses **multiple keys along with a manifest** (aka XPUB)
- Like a bank safe deposit box. Required items to access: the box number, your key, bank's key.

What is Multi-Sig

- A **single-sig wallet** is derived (usually 2026) from
- A random **number** that is **cryptographically converted** into **12/15/18/21/24** words that are then
- Cryptographically converted into a **master private and public key** and then
- Cryptographically converted into a set of **child private and public keys** that you see in your wallet

512	613-640	641-768	769-896	897-1024	1025-1152	1153-1280	1281-1408
py	513 divorce	641 exotic	769 gas	897 hybrid	1025 length	1153 mosquito	1281 par
ral	514 dizzy	642 expand	770 gasp	898 ice	1026 lens	1154 mother	1282 par
re	515 doctor	643 expect	771 gate	899 icon	1027 leopard	1155 motion	1283 par
rn	516 document	644 expire	772 gather	900 idea	1028 lesson	1156 motor	1284 par
irect	517 dog	645 explain	773 gauge	901 identity	1029 letter	1157 mountain	1285 par
st	518 doll	646 expose	774 gaze	902 idle	1030 level	1158 mouse	1286 pas
tion	519 dolphin	647 express	775 general	903 ignore	1031 liar	1159 move	1287 pat
uch	520 domain	648 extend	776 genius				
untry	521 donate	649 extra	777 genre				
uple	522 donkey	650 eye	778 gentle				
urse	523 donor	651 eyebrow	779 genuine				
usin	524 door	652 fabric	780 gesture				
ver	525 dose	653 face	781 ghost				
vyote	526 double	654 faculty	782 giant				
ack	527 dove	655 fade	783 rift				



What is Multi-Sig

- What if you cryptographically converted a bunch of single-sig wallets into one wallet?



What is Multi-Sig

- What if you cryptographically converted a bunch of single-sig wallets into one wallet?
- THAT'S MULTI-SIG!



THE END

- Thanks

What is Multi-Sig

- A Bitcoin multisig wallet is a wallet that requires **approval from more than one key** before bitcoin can be spent, making it safer than relying on a single key.
- A **shared rule**, not a shared device
- The Bitcoin **network enforces** the rule
- No company, person, or service can override it
- Keys can be **stored separately**
- Keys can be **owned by different people** or devices
- It is **similar to collecting approvals** before money moves
- Keys never need to touch each other
- One compromised key $\neq$ stolen funds



Why (should Multi-Sig be used)

- **Protection** from
 - • Theft
 - • Coercion
 - • Malware
 - • Accidental loss
 - • Environmental disaster
- Gives **time to react** instead of instant disaster
- No single key has power. **Trust is distributed**. Loss of one key $\neq$ disaster

How (is Multi-Sig used)

- **Flexibility** in design. You can design it around **your needs** by deciding where the keys are stored, who has access, when people have access, how much access is needed or available and what scenarios you want protection from.

How (is Multi-Sig used)

- Long-term personal savings
- Family or shared funds
- Inheritance planning
- Business treasuries
- Protection against theft or coercion
- Geographic risk reduction
- Device risk separation
- Collaborative custody
- Delayed or controlled spending
- Estate and contingency planning
- Large balance management
- Redundancy and recovery planning

When (should Multi-Sig be used)

- Be proactive. Ask, yourself, how comfortable are you with your current stack's storage situation when Bitcoin hits \$250k, \$1 million, \$5 million or more?



When (should Multi-Sig be used)

- Multi-Sig (savings account)
- Single-Sig (checking account)
- Lightning (cash in your wallet).

Multi-sig as a layer of security, **not a replacement**

It's time to normalize Multi-sig wallets!

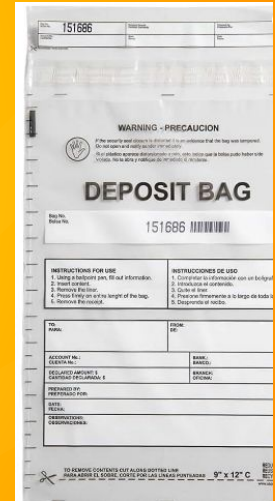
Where

(should the keys be hidden)

- Various geographic locations
- At Home
- With relatives
- Bank or private safe deposit boxes
- Buried
- Plain sight
- Fire-proof safes
- Hidey holes
- Encrypted
- Memorized
- Swap with another trusted Bitcoiner
- In your bunghole

Where (should the keys be hidden)

- You can store multiple hardware **wallets** and seed **phrase plates**, of the exact same seed phrase, in **multiple locations**.
- You can use tamper evident **deposit bags** with serial numbers.
- **Both** the hardware wallet and the seed phrase plate will **exist**. Hardware wallet and seed phrase plate hidden together or separate?
- **Mega-disaster scenarios? Worth mentioning? It is besides the point, as the proper number of multi-sig keys utilized, and hidden in the correct way, can outlive most scenarios.**



Who (should be involved)

- Your Mom

Who (should be involved)

- Pick **reliable people**
- Use **time-locks** for added safety
- Highly dependent on **your situation** and who you trust
- **Trust** isn't just in terms of theft, but **also reliability** to not compromise your key through **mishandling**.
- Trust is also that people will **keep their mouth shut**

Whom

(the keys are accessible to)

- Just you? or
- **Collaborative custody**
 - Unchained
 - Anchor Watch
 - Lawyers
 - Family
 - Friends

Which (configuration to use)

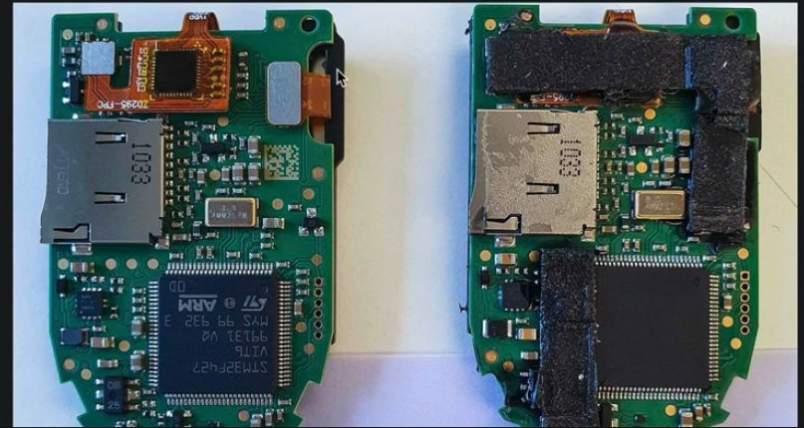
- **You choose:** How many **total** keys exist and how many are **required** to spend
- Pros and cons of different size M (required number of keys) of N (total possible keys):
 - Low M : Easier and **quicker** to access. More likely to be accessible after large loss of keys (*if M is low relative to N*).
 - High M : Safer from **collusion or theft**.
 - Low N : **Less planning** needed. **Less chance** of a compromised key.
 - High N : Less chance of **disaster** due to compromised keys. Can **incorporate** more collaborators. **More options** for access when needed.

Hard & Soft (ware) Tips

- Single **vendor** compromises.
- Problems from **supply-chain** attacks or compromised vendors are **mitigated** when using multiple different hardware wallets.
- Fake wallets

Fake Trezor Wallets Designed to Steal Bitcoin Spotted in the Wild

"It looks and feels like a Trezor wallet, but puts all your crypto-investments into the hands of criminals."



Hard & Soft (ware) Tips

- Embrace **software not produced** by the hardware manufacturer
 - Sparrow Desktop
 - Nunchuk Wallet



Hard & Soft (ware) Tips

- **Start small.** Create a **test wallet**, move Sats, test in **and OUT**. Test **access to remote keys**.
- **Test recoveries** semi-annually; **update as life changes**. Sign a **test transaction** with multiple keys, but don't broadcast.
- **Follow** all the **safety rules** of generating a single-sig wallet properly. **Use dice** for randomness (over a 100 rolls), **no cameras**, **no mics**, **no phones**. **Don't speak** out loud. Do it **alone**. Create on an **air-gapped** hardware wallet.
- Don't be afraid to use **passphrases** (13th/25th word).
- **Documentation** for your future self

Bitcoin Multi-Signature wallets for the pleb

Thank you!
Good luck!

HODL

Best Practices

Purchase Bitcoin **ethically**

Have **no single point-of-failure**

Use a **reliable software** wallet

Air-Gap

Backup your Bitcoin seed

Keep your **mouth shut**

Test your backups

Stay **humble**

Run and use a **full-node**

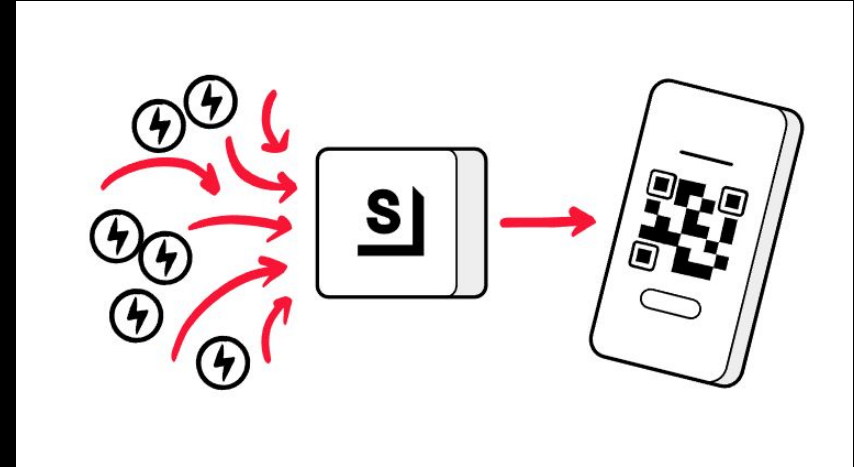
Stack sats

Manage your **UTXOs**

HODL

Ark - Second's Bark

- Bitcoin Layer 2: no channels, no liquidity management,
 - Self-custodial
 - Unilateral exit
- Server runs "rounds" every hour:
 - One on-chain tx batches hundreds of users
- Server fronts its own bitcoin as liquidity, earns fees on the service
- Tradeoff: must refresh every ~30 days or server can claim your coins
- Not replacing Lightning -- it's the on-ramp that makes Lightning accessible





Bitcoin Policy Institute 🌟
@bitcoinpolicy

X.com

🇺🇸 New Study from BPI: Frontier AI agents prefer bitcoin over stablecoins and other forms of money.

BPI tested 36 models over 9000+ conversations, and the AIs overwhelmingly chose to use Bitcoin for their economic activity.

What AI Agents Chose

Bitcoin	4,378	48.3%
Stablecoins	3,013	33.2%
Fiat & Bank Money	809	8.9%
Crypto	383	4.2%
Tokenized RWA	99	1.1%
Compute Units	86	0.9%

09:45 · 3/3/26 · 53K Views



The Bitcoin Historian ✓
@pete_rizzo_

JUST IN: YAHOO FINANCE ANALYSTS JUST SAID LIVE THAT NEARLY 70% OF AI CLAUDE BOTS ARE CHOOSING #BITCOIN OVER THE U.S. DOLLAR

“BTC IS THE MAIN CURRENCY USED BY AI AGENTS”

THE FUTURE IS HERE 🚀



12:02 PM · Mar 17, 2026 · 137K Views



John Cantrell ✓
@JohnCantrell97

I'm back.

Will be pushing on agentic bitcoin payments and building things agents will actually pay for.

First step was getting [@moneydevkit](#) working on [@Cloudflare](#).

Required building a custom [@lightningdevkit](#) node that compiles to wasm to run on cloudflare's js runtime and take advantage of their tcp, storage, and durable object apis.

Source and examples here: github.com/johncantrell97...

```
npm install mdk-cloudflare
```

Just point your agent at the repo and will be able to get it running for you.

...



John Cantrell ✓ @JohnCantrell97 · 3h
/5

Why this is a big deal:

Machines are about to be the biggest consumers of APIs. AI agents don't have credit cards. They don't want to sign up for accounts.

With MPP + Lightning, an agent just pays and gets access. The internet finally has a native payment layer.



3



3



16



555



John Cantrell ✓ @JohnCantrell97 · 3h
6/

Under the hood, mdk-cloudflare runs a Lightning node compiled to WASM inside a Cloudflare Durable Object. Your node, your keys, your bitcoin. Deployed in seconds.

```
npm install mdk-cloudflare
```



Thank You!



Hunter Beast 📌 🔗 @cryptoquick · 4h 

BIP 54 addresses threats to Bitcoin that exist right now. Poisoned blocks that take 30 minutes to validate based on legacy sigops limits are something that can happen at any time, unlike the looming threat posed by quantum computers, which still remains largely theoretical.



Jameson Lopp 🔗 📷 @lopp · 5h

Bitcoin is strong but it's not perfect; Bitcoin is vulnerable to attack. There are principled people who have been warning us for years about the need to patch Bitcoin's weaknesses, and it's long overdue for us to take action.

It's not controversial: let's activate BIP 54!

Abstract

This document proposes new consensus rules in order to fix the timewarp attack, reduce the worst case block validation time, prevent Merkle tree weaknesses, and avoid duplicate transactions without [bip-0030](#) validation.



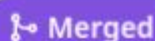
Murch @murchandamus · Mar 17



BIP446: OP_TEMPLATEHASH

BIP448: Taproot-native (Re)bindable Transactions
were published:

BIP446: OP_TEMPLATEHASH and BIP448:
Taproot-native (Re)bindable Transactions
#1974



Merged

[murchandamus](#) merged 1 commit into [bitcoin:master](#) from

[instagibbs:bip_op_templatehash](#)



1 minute ago

Jersey City Bitcoin Education Center Library & Book Sharing Program

- Bring bitcoin books you have already read and exchange for ones you haven't! Trade back the following month.
- Borrow books from current library with Sats deposit. Return next month and receive your deposit back.
- Konsensus Network provides discounts on book purchases from Bitcoin Communities. This should be available soon.